

# How are the time-series and textual data streams integrated within the model

The integration of time-series and textual data streams within a multimodal forecasting model like **MultiModalGPT-Trader** is effectively realized through a **cross-attention fusion mechanism** within a dual-tower architecture, as detailed in recent research <sup>[1]</sup>.

## Technical Breakdown of Integration

### 1. Dual-Tower Architecture

- **Time-Series Tower:** Processes raw numerical time-series data (e.g., OHLCV, RSI, MACD) using a patch-based transformer model such as **PatchTST**. The time series  $x \in \mathbb{R}^L$  (length  $L$ ) is divided into  $p$  patches, each encoded into embeddings:

$$z_{ts} = \text{PatchTST}(x) \in \mathbb{R}^{p \times d_{ts}}$$

where  $d_{ts}$  is the embedding dimension per patch. These embeddings serve as **keys (K)** and **values (V)** in the subsequent fusion step.

- **Textual Tower:** Processes financial text data (news, transcripts) using a Large Language Model (e.g., Meta Llama-3.1 8B). Token embeddings  $\{t_i\}_{i=1}^n$  are averaged to form a global semantic embedding:

$$z_{tx} = \frac{1}{n} \sum_{i=1}^n t_i \in \mathbb{R}^{d_{tx}}$$

This embedding acts as the **query (Q)** in the fusion module.

### 2. Cross-Attention Fusion Module

The core integration step uses a **cross-attention mechanism** where the text embedding queries the time-series embeddings:

$$z = \text{CrossAttention}(Q, K, V) = \text{softmax} \left( \frac{QK^\top}{\sqrt{d_k}} \right) V \in \mathbb{R}^d$$

- $Q = z_{tx}$  (text embedding)
- $K = V = z_{ts}$  (time-series patch embeddings)
- $d_k$  is the key dimension for scaling

This operation fuses textual context with time-series patterns by weighting time-series embeddings according to their relevance to the textual information.

### 3. Final Prediction Layer

The fused representation \$ z \$ is passed through feed-forward layers to generate forecasts over the desired horizon:

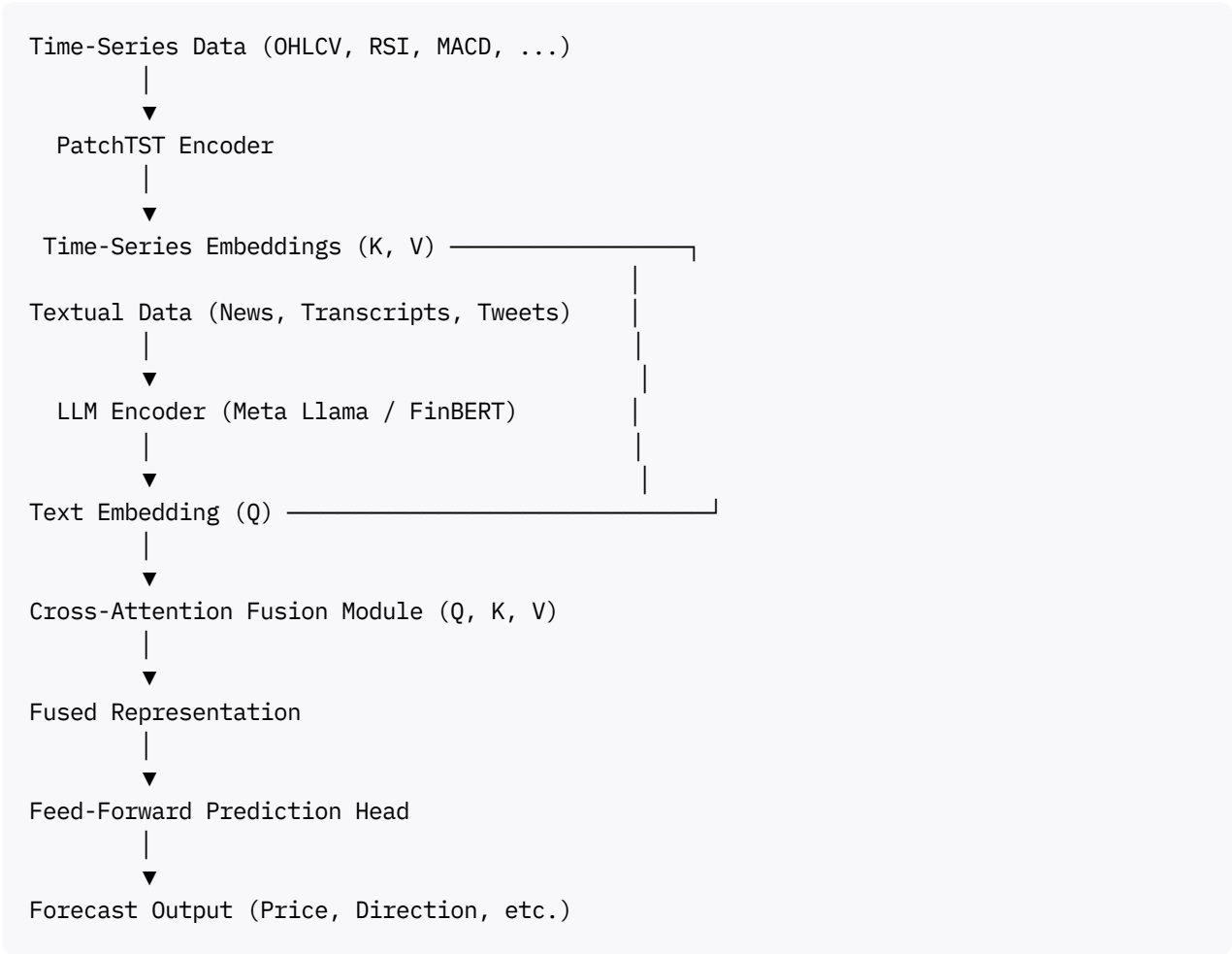
$$\hat{y} = \text{FFN}(z) \in \mathbb{R}^h$$

where \$ h \$ is the forecast horizon.

### Key Advantages of This Integration Approach

- **Context-Sensitive Fusion:** The patch-based time-series embeddings allow the static text embedding to interact differently with various temporal segments, enabling dynamic contextualization.
- **Temporal Alignment:** Although textual data is static or irregularly timed, the cross-attention mechanism aligns it adaptively with relevant time-series patches, capturing causal or correlative effects.
- **Improved Forecast Accuracy:** Experiments show that incorporating textual data via this fusion improves forecasting metrics such as MAE and WAPE, especially for longer horizons where numerical data alone may lack sufficient context.

### Summary Diagram (Conceptual)



## Reference

This integration framework is inspired by the **TextFusionHTS** model proposed by Zhou et al. (2025), which demonstrates the efficacy of cross-attention fusion between patch-based time-series embeddings and large language model text embeddings for high-dimensional time series forecasting <sup>[1]</sup>.

**In essence, the model treats the textual embedding as a query that selectively attends to different temporal patches of the time series, resulting in a rich, context-aware fused representation that improves forecasting performance.**

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1. <https://arxiv.org/html/2501.07048v1>